A

Major Project

On

PROPOUNDING FIRST ARTIFICAL INTELLIGENCE APPROACH FOR PREDICTING ROBBERY BEHAVIOUR IN AN INDOOR SECURITY CAMERA

(Submitted in partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

By

P Yuktha	(217R1A05P6)
B Deeksha	(217R1A05L7)
P Raghava Pavan	(217R1A05P8)

Under the Guidance of

Dr. RAJ KUMAR PATRA

(Professor)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CMR TECHNICAL CAMPUS

UGC AUTONOMOUS

(Accredited by NAAC, NBA, Permanently Affiliated to JNTUH, Approved by AICTE, New Delhi)

Recognized Under Section 2(f) & 12(B) of the UGCAct.1956,

Kandlakoya (V), Medchal Road, Hyderabad-501401.

April, 2025.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the project entitled "**PROPOUNDING FIRST ARTIFICAL INTELLIGENCE APPROACH FOR PREDICTING ROBBERY BEHAVIOUR IN AN INDOOR SECURITY CAMERA**" being submitted by **P. YUKTHA (217R1A05P6), B. DEEKSHA (217R1A05L7) & P. RAGHAVA PAVAN (217R1A05P8)** in partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2024-25.

The results embodied in this project have not been submitted to any other University or Institute for the award of any degree or diploma.

Dr. Raj Kumar Patra **Professor** **INTERNAL GUIDE**	**Dr. Nuthanakanti Bhaskar** **HoD**
Dr. A. Raji Reddy **DIRECTOR**	**Signature of External Examiner**

Submitted for viva voice Examination held on ____________________

ACKNOWLEDGEMENT

We take this opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project, we take this opportunity to express our profound gratitude and deep regard to our guide **Dr. Raj Kumar Patra,** Professor for his exemplary guidance, monitoring and constant encouragement throughout the project work. The blessing, help and guidance given by him shall carry us a long way in the journey of life on which we are about to embark.

We take this opportunity to extend our heartfelt appreciation to the Project Review Committee (PRC) Coordinators—**Dr. K. Maheswari, Dr. J. Narasimharao, Ms. K. Shilpa, and Mr. K. Ranjith Reddy**—for their unwavering support, insightful guidance, and valuable inputs, which played a crucial role in steering this project through its various stages.

Our sincere appreciation also goes to **Dr. Nuthanakanti Bhaskar**, Head, for his encouragement and continuous support in ensuring the successful completion of our project.

We are deeply grateful to **Dr. A. Raji Reddy**, Director, for his cooperation throughout the course of this project. Additionally, we extend our profound gratitude to Sri. **Ch. Gopal Reddy**, Chairman, Smt. **C. Vasantha Latha**, Secretary and Sri. **C. Abhinav Reddy**, Vice-Chairman, for fostering an excellent infrastructure and a conducive learning environment that greatly contributed to our progress.

We also acknowledge and appreciate the guidance and assistance provided by the faculty and staff of **CMR Technical Campus**, whose contributions have been invaluable in bringing this project to fruition.

Lastly, we sincerely thank our families for their unwavering support and encouragement. We also extend our gratitude to the teaching and non-teaching staff of CMR Technical Campus for their guidance and assistance. Their contributions, along with the support of everyone who helped directly or indirectly, have been invaluable in the successful completion of this project.

P Yuktha (217R1A05P6)

B Deeksha (217R1A05L8)

Raghava Pavan (217R1A05P8)

VISION AND MISSION

INSTITUTE VISION:

To Impart quality education in serene atmosphere thus strive for excellence in Technology and Research.

INSTITUTE MISSION:

1. To create state of art facilities for effective Teaching- Learning Process.
2. Pursue and Disseminate Knowledge based research to meet the needs of Industry & Society.
3. Infuse Professional, Ethical and Societal values among Learning Community.

DEPARTMENT VISION:

To provide quality education and a conducive learning environment in computer engineering that foster critical thinking, creativity, and practical problem-solving skills.

DEPARTMENT MISSION:

1. To educate the students in fundamental principles of computing and induce the skills needed to solve practical problems.
2. To provide State-of-the-art computing laboratory facilities to promote industry institute interaction to enhance student's practical knowledge.
3. To inculcate self-learning abilities, team spirit, and professional ethics among the students to serve society.

ABSTRACT

Crime prediction in video-surveillance systems is required to prevent incident and protect assets. This project presents the first artificial intelligence approach for predicting Robbery Behavior Potential (RBP) using indoor video surveillance systems, aimed at preventing incidents and protecting assets. Our method integrates three key detection modules: head cover detection, crowd analysis, and loitering detection, each designed to trigger timely interventions. We retrain the YOLOv5 model on a manually annotated dataset for the first two modules, while the loitering detection module innovatively leverages the DeepSORT algorithm, supported by a fuzzy inference system that applies expert rules to assess robbery potential. Despite challenges such as varied robber behavior, camera angles, and low video resolution, our experiments on real-world surveillance footage achieved an F1-score of 0.537. By establishing a threshold value for RBP, we benchmarked our method against existing robbery detection techniques, achieving a significant improvement with an F1-score of 0.607. Our findings suggest that this approach can enhance situational awareness for human operators and optimize the management of surveillance cameras, ultimately contributing to a reduction in robbery incidents.

LIST OF FIGURES

LIST OF TABLES

TABLE OF CONTENTS

1. INTRODUCTION

1. INTRODUCTION

Surveillance cameras play a crucial role in public safety by monitoring activities and assisting in crime detection. However, manual monitoring is prone to fatigue and human error, making automated anomaly detection essential. This study introduces a novel approach to predicting robbery behavior (RBP) in indoor spaces using three key modules: head cover detection, crowd analysis, and loitering detection. By extracting visual features and leveraging a fuzzy inference machine, the system enhances prediction accuracy and overcomes challenges such as low-resolution surveillance footage. Unlike deep learning methods, fuzzy logic provides adaptability, allowing for nuanced decision-making based on expert knowledge.

The proposed algorithm employs a manually annotated dataset, optimizing features for effective robbery behavior prediction. The loitering detection module, powered by Deep SORT tracking, assigns points based on movement analysis using Euclidean distance calculations. Additionally, a fuzzy inference machine refines predictions by integrating expert-defined rules for robbery scenarios. The method enhances crime prevention by predicting potential threats and alerting authorities in real-time. Future work will focus on improving accuracy through advanced feature extraction and expanding the dataset to accommodate diverse robbery scenarios.

1.1 PROJECT PURPOSE

The purpose of this project is to develop an intelligent system for predicting robbery behaviour (RBP) in indoor environments using surveillance videos. Traditional security systems rely on human monitoring, which can be inefficient due to fatigue and subjectivity. This project aims to automate the detection of suspicious activities by analysing key behavioural patterns such as head covering, crowd density, and loitering. By integrating machine learning and fuzzy inference techniques, the system enhances real-time crime prediction, improving response times and preventing potential threats before they escalate.

By leveraging deep learning-based tracking and fuzzy logic for decision-making, this project provides a more adaptable and accurate method for identifying potential robbery scenarios. The system is designed to work effectively with low-resolution video feeds and single-camera setups, making it suitable for a wide range of surveillance applications. Ultimately, this project contributes to public safety by assisting security personnel in identifying high-risk situations, reducing financial losses, and preventing violence in commercial spaces.

1.2 PROJECT FEATURES

This project incorporates several key features to improve the accuracy and efficiency of robbery prediction and detection.

The project integrates YOLOv5, a state-of-the-art object detection algorithm, to identify key visual elements related to robbery behaviour prediction (RBP). YOLOv5 is used to detect head coverings such as masks, helmets, or hoods, which are commonly used by offenders to conceal their identities. Additionally, it plays a crucial role in crowd analysis by identifying the number of individuals present in a scene, helping to determine if a location is unusually crowded. Its real-time processing capability ensures that even low-resolution surveillance footage can be effectively analysed, making it suitable for various security applications.

To enhance behavioural analysis, the Euclidean Distance algorithm is implemented for loitering detection. It calculates the movement patterns of individuals by measuring the distance they travel within a specific period. If a person remains within a confined space for an extended time without a clear purpose, the system assigns a loitering score based on movement data. This approach helps detect suspicious activity, as offenders often linger around potential targets before committing a crime. The integration of Euclidean Distance with tracking methods allows the system to differentiate between normal and potentially criminal behaviour.

For improved tracking and identity consistency across frames, Deep SORT (Deep Simple Online Real-time Tracker) is employed. This algorithm ensures that detected individuals are continuously tracked, preventing identity mismatches in subsequent frames. By maintaining a consistent identity for each person, Deep SORT enhances loitering detection accuracy and reduces false positives. The combination of YOLOv5 for detection, Euclidean Distance for movement analysis, and Deep SORT for tracking results in a robust, intelligent system capable of predicting robbery behaviour in real-time, ultimately improving security measures in commercial and public spaces.

By integrating these features, a fuzzy inference machine integrates expert knowledge to interpret outputs from all three modules, enhancing predictive accuracy.

2 LITERATURE SURVEY

2. LITERATURE SURVEY

Anomalies are infrequent observations, events or behaviors which are suspicious because they are significantly different from normal patterns. Crime is a kind of an anomaly which is any behavior deviating from a normal activity. One could say that the proliferation usage of CCTV has been because of increasing crimes in public places. Crime can be predicted according to suspicious behavior detection. prediction needs defective, vague and unsure information. Our proposed approach concentrates on RBP prediction in indoor places. Robbery is a kind of crime and the proposed algorithm needs loitering, crowd and head cover detection. One important concept of our algorithm is providing a generic RBP prediction framework which is not addressed in any other paper. In this section, we will discuss about some related works relevant for suspicious behavior detection or prediction, crime detection or prediction and articles concerning with the loitering or head cover detection.

Elhamod and Levine proposed a semantics-based suspicious behavior recognition algorithm based on object tracking by blob matching with color histograms and spatial information, for updating objects intended in each frame. For blob and objects similarity specification, intersection of histogram's value is calculated and compared with the defined threshold. Next it assigns appropriate classes contains people for animated and objects for inanimated things. By calculating their 3D motion features and recording it in the form of historical records, behaviors are semantically determined. detected suspicious behaviors include: abandoned luggage by background subtraction methods, fainting by comparing assumed 2D and actual 3D location of person's feet and also head coordinates of that person, fighting by computing merge, split and simultaneously movement of blob's centroid and eventually loitering by aggregating presence time of a person in an area.

Ishikawa and Zin, introduced an automated normal system for questionable pedestrian detection by loitering detection. According to , a questionable person walks, stops and goes around the location repeatedly for a long time with enhancement of direction changing number. His distance value is greater than the normal person and changing in acceleration is so much. To implement these features, divides the video frames into 25 blocks and counts the frequency of block numbers which feet of person are in that location. if this frequency was more than threshold value, that person descent as suspicious pedestrian. To compute changing of direction, it calculates angles of moving direction. Computing of distance and acceleration changing extracts all needed features. Finally, a decision fusion process detects suspicious pedestrians by aggregating the scores of each step.

Rajapakshe et al. presented an E-police system which contains two components: video surveillance monitoring system and crime prediction. To detect suspicious behaviors such as violent and vandalism, uses human activity recognition methods and classifies them into normal and abnormal categories. They use Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) for feature extraction and detect suspicious human who conceal their faces. crime prediction process of is based on public information resources for place and time of crime occurrence prediction with the help of classification algorithms such as SVM and Decision tree.

Arroyo et al. proposed an expert real-time suspicious behaviors detection system in shopping malls. They locate foreground objects by an image segmentation and background subtraction algorithm. Next, a blob fusion algorithm is used to gather the blobs of each segmented parts to detect human. A tracking algorithm is used with the help of a new two-step method: a) using a Kalman filter for detection and tracking human, b) SVM kernels for occlusion management. Then, the obtained trajectories of people are used to analyze human behaviors. The entrance or existence alarm is for the time that too many people enter or a person runs away and it is detected by trajectory analyzing. Moreover, specific risk areas are interiors with more expensive articles and chosen by the human security officers. Loitering of people is evaluated according to their trajectories and the length of time they present in those zones. If the time be more than 30 seconds, which has specified by security experts, their system gives off an alarm. They mounted a camera on the cash desk to protect it. If someone loiters around it, and no shop personnel attended in the cash desk zone, an alarm gives off. Additionally, evaluation process is done on a naturalistic dataset they provided by multi cameras located on entrance, interior and cash-desk of a shop.

Bouma et al. focused on the automatic pickpocket behavior early detection by three steps: detection and tracking pedestrians using multicamera, feature extraction adopted to pickpocket scenario and diagnosis of pickpocket. Based on an expert knowledge, the scenario of pickpocket assumed in includes seven main phases: surroundings observation, looking for a proper opportunity, communicating to accomplice, besieging target person, snatching a stuff, handing it over and evacuating the scene. To implement this scenario, they extract some related features including speed of walking, changing of orientation and crowd merging and splitting. Finally, a Fisher Linear Discriminant Classifier (LDC) used to recognize pickpockets. They used video dataset which has been played by some actors as pickpockets and victims.

Selvi et al. determined enhanced CNN (ECNN) algorithm to detect suspicious actions like shooting and stealing. Their ECNN method have input layer to feed the convolution3D layer for feature extraction. Next, methods of set threshold for LeakyReLU layer for detection of normal or suspicious action.

Roy proposed a snatch theft detection model by using Gaussian Mixture Model (GMM) with regards to scenarios of snatch theft occurrence. Reference use HOF and MBH as feature descriptors to extract dense trajectory feature sets and represented as action-vector. Next, they train GMM by these vectors for snatch theft detection.

Kaur et al. presented a face mask detection algorithm by retraining a CNN with 3832 face dataset images consisting with or without mask. A face mask detection system is developed by retraining YOLOV5 with 685 images that consists of images of people from two categories that are with and without face masks. Proposed method of training their model with different number of epochs and the optimal epoch number they reported was 300. Face mask detection model proposed in, retrained MobilenetV2 with the help of data augmentation methods. authors of proposed a method which detects the face mask with the help of deep Neural Networks. They fine-tuned the last layer of RESNET50 by adding five new layers and retrained it with 25876 images including with or without mask faces. Chowdary et al. added 5 more layers to InceptionV3 to finetune it for mask detection. Singh et al. trained both YOLOv3 and faster R-CNN with the images including human with or without face mask. Huang et al. proposed a method to detect helmet with an improved YOLOV3 with the change in feature map size of it for small object detection. Zhou et al. retrained YOLOV5 with 6045 images including people in two categories: with or without helmet for helmet detection. A retrained Single Shot Detectors (SSD) MobileNet V2 deep network used in for helmet detection. YoloV5 has small size and it is faster to train. Furthermore, it has ability of small object detection. Some loitering detection methods are discussed in ensuing part.

Authors of proposed a loitering detection method in which person detected by a YOLOV3 algorithm and tracked by a DeepSORT algorithm. Nayak et al. detected loitering of individual by computing time duration and displacement of them in comparing with a threshold they defined. Nam proposed a loitering detection algorithm based on pedestrian detection by blob detection and comparing spatial and temporal information with a threshold to detect loitering person. The time threshold defined in calculated with regards to time staying in the place and the frame rate. Consequently, our proposed algorithm originated from behavior detection systems like proposed systems of and which used scenarios define suspicious behaviors and different steps accomplished to detect these behaviors. We use a scenario of robbery behavior occurrence to implement our innovative method. In our proposed method, we elaborate on method of and we use DeepSORT algorithm for tracking each person. We also use Euclidean distance for loitering detection based on defining a threshold for distance value. A YoloV5-based algorithm is retrained for mask and helmet detection for head cover detection module.

A quick summary of all these methods is provided in Table 1. Furthermore, none of the methods defined above, have pondered the RBP prediction with purpose of preventing its occurrence. The potential for robbery behavior can only be calculated in videos with analyzing possibility for a period of time before any force, threat or display of a weapon occurs. That is, the suspicious behavior of the robber can be assessed. Common scenarios of robbery behavior, is choosing a poorly attended stores, covering head to not be recognized, and loitering for an opportunity to threat with a weapon and use force. In our proposed approach, number of human and his head cover is detected using the retrained YOLOV5. Next, the human is tracked using DeepSORT for our novel loitering calculation method which is based on Euclidean distance calculation and our defined displacement thresholds. A human need for inferring a video and decide about RBP of each frame. Fuzzy Logic looks at the world in ambiguous terms, in much the same way that our brain takes in information, then responds with precise actions. The human brain can reason with uncertainties, vagueness, and judgments. Computers can only manipulate precise valuations. Fuzzy logic is an attempt to combine the two techniques. Due to the imprecise terms resulted from three modules, the fuzzy inference machine can, like the human mind, decide on the potential of robbery behavior. So, a fuzzy inference machine inferences RBP.

Reference	Proposed Scheme	Implemented Methods
[18]	suspicious persons detection system by loitering detection	To implement this system, they calculate traveled distance, acceleration and direction changes. Finally, they compare calculated parameters with a threshold they have defined to decide if the person is suspicious.
[22]	An expert video-surveillance system is utilized for detection of real-time suspicious behaviors like loitering around interior or cash desk and running, in shopping malls.	The proposed system detects humans by blob fusion algorithm and track them by Kalman filter and SVM kernels. Analysing of behaviours done by obtained trajectories.
[27]	Real-time detection of suspicious behaviors like loitering in public transport areas	The proposed solution recognize behaviour semantically based on object tracking and assigning a label as object or human for each one. Next, they calculate 3D motion features to detect suspicious behavior
[28]	Suspicious behavior detection and spatio-temporal crime prediction	CNN and RNN methods are used for feature extraction to recognize human activity and detect suspicious behavior. they also use public information resources belong to place and time of crime occurrence for crime prediction.
[29]	Automatic early detection of a suspecious behavior with track-based features in a shopping mall. they detect behaviour of pickpocket by it's scenario implementation.	Feature extraction and Fisher linear discriminant classifier is used to early detect pickpocket. Their features are based on an expert knowledge including: speed of walking, changing of orientation and crowd merging and splitting.
[30]	Suspicious action detection	An ECNN algorithm proposed for suspicious action detection.
[31]	Snatch theft detection model	Their model uses GMM based on common theft scenarios. it also uses HOF and MBH to extract dense trajectory feature sets and create action-vectors to train GMM. .
[32] − [37]	Deep learning algorithms are used for face mask detection.	Retraining CNN, YOLOV5, MobilenetV2, RESNET50, InceptionV3, YOLOV3 and faster R-CNN with two-class images: with or without mask.
[38] − [40]	Approaches based on deep learning algorithms are used to detect helmet.	Proposed solutions detect helmet by retraining YOLOV3, YOLOV5 and SSD MobileNet V2.
[41]	Loitering detection method based on human detection and track.	A YOLOV3 based detector is used for human detection and a DeepSORT algorithm for tracking him. By comparing time duration and displacement of individuals with a defined threshold, loitering is detected.
[42]	Spatio-temporal based detection system for loitering.	A YOLOV3 based detector is used for human detection and a DeepSORT algorithm for tracking him. By comparing time duration and displacement of individuals with a defined threshold, loitering is detected.

2.1 REVIEW OF RELATED WORK

The detection and classification of inappropriate content in videos have been extensively studied in the fields of computer vision, deep learning, and content moderation. Several approaches have been proposed to tackle the problem, ranging from rule-based filtering techniques to advanced deep learning models. This review discusses previous research and existing methodologies, highlighting their strengths and limitations.

1. Real-time automatic detection of vandalism behavior in video sequences

A technique for the real-time identification of vandalism in video sequences is presented in this research. The suggested approach uses a single camera and robustly extracts a series of high-level events that precede vandalism in order to identify it without the need for object identification. When an item, such a pay phone or a sign, enters the scene and modifies anything without permission inside a designated vandal-sable region, it is considered vandalism. Our findings demonstrate that the suggested approach is reliable for identifying graffiti or vandalism in surveillance video sequences, having been tested .

2. Automatic video-based human motion analyzer for consumer surveillance system

Automatic low-cost video surveillance is progressively becoming available for consumer applications as a result of ongoing advancements in video-analysis technology. Video surveillance may make it easier to regulate equipment use and home entry while also enhancing occupant safety. In this work, we examine a versatile framework for semantic analysis of human behavior from consumer camera captured monocular surveillance footage. Semantic analysis of human actions and events in video sequences is made easier by successful trajectory estimation and human-body modeling. The adoption of a 3-D reconstruction technique for scene interpretation is an additional feature that allows for the analysis of human behavior from various perspectives. Four processing levels make up the entire framework: (1) a preprocessing level that includes multiple-person detection and background modeling; (2) an object-based level that performs posture classification and trajectory estimation; (3) an event-based level that performs semantic analysis; and (4) a visualization level that includes camera calibration and 3-D scene reconstruction. After evaluation, our suggested framework demonstrated its high quality (86% posture classification accuracy and 90% event classification accuracy) and efficacy, achieving performance close to real-time (6-8 frames/second).

3. Real-world anomaly detection in surveillance videos

A wide range of realistic abnormalities may be captured in surveillance footage. In this work, we suggest using both normal and anomalous films to learn abnormalities. We suggest using weakly labeled training videos—that is, training labels (normal or anomalous) are at the video-level rather than the clip-level—to learn anomaly through the deep multiple instance ranking framework in order to avoid annotating the anomalous segments or clips in training videos, which takes a lot of time. Our method automatically learns a deep anomaly ranking model that predicts high anomaly scores or anomalous video segments by treating normal and anomalous films as bags and video segments as instances in multiple instance learning (MIL). In order to properly localize anomaly during training, we also add temporal smoothness and sparsity restrictions to the ranking loss function. We also provide a brand-new, extensive dataset with 128 hours of films. It is made up of 1900 uncut real-world surveillance movies that include 13 actual anomalies, including robberies, fights, car crashes, and burglaries, in addition to everyday activity. There are two uses for this dataset. The first step is general anomaly identification, which takes into account all regular activity in one group and all abnormalities in another. Second, for identifying each of the thirteen unusual acts. Our experimental findings demonstrate that, in comparison to state-of-the-art techniques, our MIL method for anomaly identification significantly improves anomaly detection performance. We provide the findings from a number of recent deep learning baselines on the identification of aberrant behavior. These baselines' poor recognition performance indicates how difficult our dataset is and provide more room for further research. The dataset is accessible.

4. Automated visual surveillance in realistic scenarios

In this post, we introduce Knight, an automated surveillance system used in a range of real-world situations, from law enforcement to railroad security. We also go over the difficulties in creating surveillance systems, show some of the ways that Knight has addressed these difficulties, and assess how well Knight performs in unrestricted settings.

5. Helmet detection based on improved YOLO V3 deep model

Wearing a helmet is crucial for workers' safety in factories and construction sites. It might be challenging for businesses to keep an eye on how to alert, identify, and certify employees "whether or not the helmet is worn." This paper uses the advantage of Dense net in model parameters and technical cost to replace the backbone of the YOLO V3 network for feature extraction , creating the so called YOLO - Dense backbone convolutional neural network. It is based on the YOLO V3 full regression deep neural network architecture. According to the test findings, the enhanced model can handle scenarios when the helmet is partly obscured, soiled, or has a large number of targets with poor picture quality. With the same detection rate, the enhanced algorithm's detection accuracy in the test set rose by 2.44% when compared to the conventional YOLO V3. The development of this model has significant applications in enhancing helmet detection and guaranteeing secure construction.

This review highlights the evolution of content moderation techniques, emphasizing the shift from rule-based and machine learning approaches to deep learning-based models. The proposed methodology aims to address the limitations of previous research by offering a highly accurate, scalable, and efficient solution for inappropriate behaviour detection in a indoor security camera.

2.2 DEFINITION OF PROBLEM STATEMENT

This project aims to develop an AI-driven system that predicts potential robbery behavior using indoor security camera footage. By leveraging machine learning and deep learning, it analyzes human movements, facial expressions, and body posture to detect suspicious activities. The system processes real-time video feeds to identify behavioral patterns that may indicate a possible crime. Providing early warnings to security personnel enables timely intervention and enhances overall security measures. This approach bridges the gap between traditional surveillance methods and AI-powered proactive security solutions. Ultimately, the goal is to improve detection accuracy while minimizing false alarms.

2.3 EXISTING SYSTEM

The existing systems for suspicious behavior detection and crime prediction utilize a range of methodologies to identify and analyze anomalies captured through video surveillance. These systems are designed to recognize behaviors that deviate from established norms, which is crucial in detecting potential criminal activities such as robbery. For instance, The existing systems for predicting Robbery Behavior Potential (RBP) in indoor environments often rely on traditional algorithms such as basic motion detection and background subtraction methods for activity monitoring. These algorithms, Gaussian Mixture Model (GMM) for background modeling or threshold-based object tracking techniques, are limited in their ability to detect complex behaviors. Simple blob detection or region-based tracking is used for crowd movement analysis, but these lack the precision to classify individuals or detect nuanced patterns such as loitering. Additionally, algorithms like Optical Flow may be applied for movement tracking, but they are computationally intensive and fail to provide high-level behavioral insights. Despite these advancements, existing systems face significant challenges, including the complexity of data interpretation, which can hinder accurate detection, issues with data availability that can lead to unreliable predictions, and the potential for incorrect labeling within training datasets, which undermines the overall efficacy of machine learning models. Without advanced frameworks like object detection models (e.g., YOLO) or tracking algorithms (e.g., DeepSORT), these traditional approaches fall short in addressing dynamic and complex scenarios.

Limitations of Existing System

Despite improvements in video content classification, the existing system suffers from the following challenges:

- Complexity of data interpretation can affect model accuracy: In the context of predicting robbery behavior, human actions can be highly complex and vary based on context, lighting, and occlusions in indoor environments. If the model misinterprets harmless actions as suspicious or fails to recognize actual threats, its accuracy and reliability will be compromised.
- Insufficient data availability can lead to poor prediction performance: Training an AI model for robbery prediction requires a diverse dataset containing various robbery and non-robbery scenarios. If the dataset lacks sufficient real-world examples, the model may struggle to generalize and fail to detect unusual but genuine threats accurately.

- Incorrect labelling in training datasets can result in inaccurate predictions: If suspicious and non-suspicious behaviors are not correctly labeled in the dataset, the model may learn incorrect associations. This could lead to false alarms or, worse, the failure to detect actual threats, reducing the system's effectiveness in real-world security applications.

2.4 PROPOSED SYSTEM

The proposed system aims to revolutionize the prediction of Robbery Behavior Potential (RBP) in indoor environments through a comprehensive algorithm grounded in three key modules: head cover detection, crowd analysis, and loitering detection. We retrain the YOLOv5 model on a manually annotated dataset for the first two modules, system effectively distinguishes between individuals with and without head coverings, allowing for nuanced crowd counting based on these classifications. The loitering detection module employs the DeepSORT algorithm to track individual's movements, calculating loitering times through Euclidean distance analysis, thus offering a detailed understanding of individual behavior patterns. This innovative approach to defining and measuring loitering enhances the system's ability to identify suspicious activities. A significant advancement is the integration of a fuzzy inference machine, which applies expert knowledge and optimized rules to interpret the outputs of all three detection modules, allowing for robust predictions regarding robbery potential. By addressing the limitations of existing systems, the proposed framework not only improves detection accuracy but also enhances situational awareness for security personnel, thereby reducing the likelihood of robbery incidents.

Advantages of the Proposed System:

The proposed system significantly improves upon the existing approaches by addressing key limitations:

- Head Cover Detection: Effectively identifies individuals attempting to conceal their identities, a common tactic in robbery.

- Crowd Detection: Provides accurate assessments of the number of people present, helping to identify unusual congregations that may signal potential threats.
- Advanced Loitering Detection: Uses the DeepSORT algorithm for precise tracking and loitering calculations based on movement patterns, offering a detailed view of suspicious behaviour.
- Tailored Dataset: The dataset is specifically created to reflect scenarios relevant to robbery, enhancing model training and prediction accuracy.
- Fuzzy Inference Machine: Incorporates expert knowledge into the prediction process, improving the reliability of robbery potential assessments and enabling informed decision-making for security personnel.
- Real-Time Monitoring: Designed for immediate processing and response, the system enhances situational awareness and allows for prompt action in potentially threatening situations.

2.5 OBJECTIVES

- Develop an AI-based surveillance system -That analyzes indoor security camera footage to predict potential robbery behavior in real-time.
- Utilize machine learning and deep learning techniques - To detect suspicious human movements, facial expressions, and body posture indicative of criminal intent.
- Enhance security measures by providing early warnings - To security personnel, enabling timely intervention and crime prevention.
- Improve detection accuracy while minimizing false alarms - By refining data interpretation and reducing biases in model predictions.
- Bridge the gap between traditional surveillance and AI-powered security solutions - To create a more proactive and intelligent threat detection system.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- Processor : Intel Core i3
- Hard disk : 20GB.
- RAM : 4GB.

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

- Operating system : Windows 10
- Language : Python
- Back-End : Django-ORM
- Frame Work : Tkinter

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project, encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, enhancing collaboration and reducing complexity.

3.1 PROJECT ARCHITECTURE

This project architecture shows the procedure followed for the prediction and detection of robbery behaviour, starting from input to final prediction.

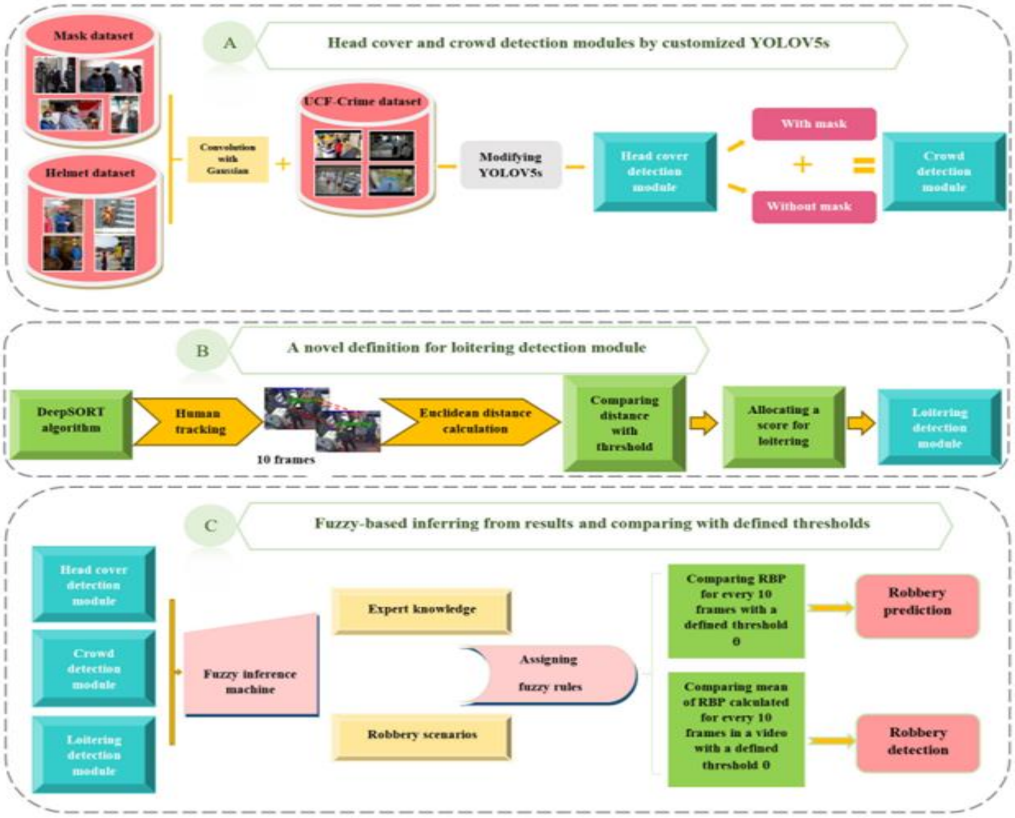

Figure 3.1: Project Architecture of Propounding First Artificial Intelligence Approach for predicting Robbery Behavior In an Indoor Security Camera.

3.2 DESCRIPTION

Data Collection: Collection of indoor security footage from various sources. Annotation and labeling of suspicious and non-suspicious activities.

Preprocessing: Raw data undergo preprocessing. Data augmentation, noise reduction, and normalization techniques. Splitting data into training, validation, and test sets.

Object Detection and Person Tracking: Object detection models like YOLO and Faster R-CNN identify humans and objects (e.g., weapons, masks, bags) in security footage. SORT and DeepSORT track individuals across frames, maintaining unique IDs for movement analysis. Feature extraction, including bounding box coordinates, speed, and trajectory, helps detect suspicious behavior patterns.

Behavioral Analysis and Anomaly Detection: Human posture and movement patterns are extracted to analyze body positions and gestures in real-time. Suspicious activities such as loitering, sudden movements, and unusual hand gestures are detected based on predefined behavioral patterns. Anomaly detection models SVM help identify deviations from normal behavior, enabling early prediction of potential robbery attempts.

Predictive Model for Robbery Behaviour: Training machine learning/deep learning models (e.g., CNNs) to predict robbery intent. Feature selection based on behavioral cues (e.g., body language, proximity to exits, sudden acceleration).Model evaluation using precision, recall, F1-score.

3.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows within a system, showcasing its processes, data stores, and external entities. It is a vital tool in system analysis and design, helping stakeholders visualize the movement of information, identify inefficiencies, and optimize workflows.

A Data Flow Diagram comprises Four primary elements:

- External Entities: Represent sources or destinations of data outside the system.
- Processes: Indicate transformations or operations performed on data.
- Data Flows: Depict the movement of data between components.
- Data Stores: Represent where data is stored within the system.

These components are represented using standardized symbols, such as circles for processes, arrows for data flows, rectangles for external entities, and open-ended rectangles for data stores.

Benefits:

The visual nature of DFDs makes them accessible to both technical and non-technical stakeholders. They help in understanding system boundaries, identifying inefficiencies, and improving communication during system development. Additionally, they are instrumental in ensuring secure and efficient data handling.

Applications:

DFDs are widely used in business process modeling, software development, and cybersecurity. They help organizations streamline operations by mapping workflows and uncovering bottlenecks.

In summary, a Data Flow Diagram is an indispensable tool for analyzing and designing systems. Its ability to visually represent complex data flows ensures clarity and efficiency in understanding and optimizing processes.

Levels of DFD:

DFDs are structured hierarchically:

- Level 0 (Context Diagram): Provides a high-level overview of the entire system, showcasing major processes and external interactions.
- Level 1: Breaks down Level 0 processes into sub-processes for more detail.
- Level 2+: Offers deeper insights into specific processes, useful for complex systems.

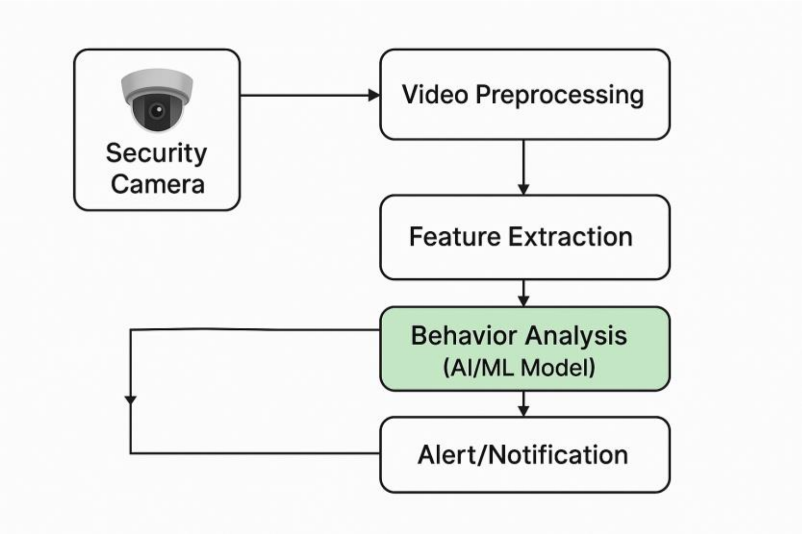

Figure 3.2: Dataflow Diagram of Propounding First Artificial Intelligence Approach for Predicting Robbery Behavior In an Indoor Security Camera.

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of a project involves executing the planned strategies and tasks. It requires meticulous coordination, resource allocation, and monitoring to ensure that objectives are met efficiently. Effective implementation is crucial for achieving project goals and delivering expected outcomes within the set timeline and budget constraints.

4.1 ALGORITHMS USED

Decision Tree Classifiers

Decision tree classifiers are used successfully in many diverse areas. Their most important feature is the capability of capturing descriptive decision making knowledge from the supplied data. Decision tree can be generated from training sets. The procedure for such generation based on the set of objects (S), each belonging to one of the classes C1, C2, …, Ck is as follows:

- Step 1. If all the objects in S belong to the same class, for example Ci, the decision tree for S consists of a leaf labeled with this class

- Step 2. Otherwise, let T be some test with possible outcomes O1, O2,…, On. Each object in S has one outcome for T so the test partitions S into subsets S1, S2,… Sn where each object in Si has outcome Oi for T. T becomes the root of the decision tree and for each outcome Oi we build a subsidiary decision tree by invoking the same procedure recursively on the set Si.

Gradient Boosting

Gradient boosting is a machine learning technique used in regression and classification tasks, among others. It gives a prediction model in the form of an ensemble of weak prediction models, which are typically decision trees.[1][2] When a decision tree is the weak learner, the resulting algorithm is called gradient-boosted trees; it usually outperforms random forest. A gradient-boosted trees model is built in a stage-wise fashion as in other boosting methods, but it generalizes the other methods by allowing optimization of an arbitrary differentiable loss function.

Advantages of Gradient Boosting Models

- High Predictive Accuracy – Gradient Boosting reduces bias and variance by combining multiple weak learners (usually decision trees), leading to superior predictive performance compared to many other models.
- Handles Various Data Types – It works well with both numerical and categorical data, making it versatile for different machine-learning problems like classification and regression.
- Feature Importance & Interpretability – It provides insights into feature importance, helping in understanding which variables contribute the most to predictions.

Disadvantages of Gradient Boosting Models

- Computationally Expensive – Gradient Boosting is slower to train compared to simpler models like decision trees or random forests, especially on large datasets.
- Prone to Overfitting – If not properly tuned, it can overfit the training data, leading to poor generalization on unseen data.
- Requires Careful Hyperparameter Tuning – It has several hyperparameters (like learning rate, number of estimators, and tree depth) that must be fine-tuned for optimal performance, making it complex to optimize.

K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) algorithm is a simple yet powerful classification method that relies on a similarity measure to classify new data points. It is a non-parametric and lazy learning algorithm, meaning it does not make any assumptions about the data distribution and does not build a model during training. Instead, KNN defers computation until a test example is provided. When a new data point needs to be classified, the algorithm identifies its K-nearest neighbors from the training data based on a chosen distance metric (such as Euclidean distance). The class of the new data point is then determined by the majority vote of its nearest neighbors, making KNN an intuitive and effective method for classification tasks.

Example

- Training dataset consists of k-closest examples in feature space
- Feature space means, space with categorization variables (non-metric variables)
- Learning based on instances, and thus also works lazily because instance close to the input vector for test or prediction may take time to occur in the training dataset

Logistic regression Classifiers

Logistic regression analysis studies the association between a categorical dependent variable and a set of independent (explanatory) variables. The name logistic regression is used when the dependent variable has only two values, such as 0 and 1 or Yes and No. The name multinomial logistic regression is usually reserved for the case when the dependent variable has three or more unique values, such as Married, Single, Divorced, or Widowed. Although the type of data used for the dependent variable is different from that of multiple regression, the practical use of the procedure is similar.

Logistic regression competes with discriminant analysis as a method for analyzing categorical-response variables. Many statisticians feel that logistic regression is more versatile and better suited for modeling most situations than is discriminant analysis. This is because logistic regression does not assume that the independent variables are normally distributed, as discriminant analysis does.

This program computes binary logistic regression and multinomial logistic regression on both numeric and categorical independent variables. It reports on the regression equation as well as the goodness of fit, odds ratios, confidence limits, likelihood, and deviance. It performs a comprehensive residual analysis including diagnostic residual reports and plots. It can perform an independent variable subset selection search, looking for the best regression model with the fewest independent variables. It provides confidence intervals on predicted values and provides ROC curves to help determine the best cutoff point for classification. It allows you to validate your results by automatically classifying rows that are not used during the analysis.

Naïve Bayes

The naive bayes approach is a supervised learning method which is based on a simplistic hypothesis: it assumes that the presence (or absence) of a particular feature of a class is unrelated to the presence (or absence) of any other feature. Yet, despite this, it appears robust and efficient. Its performance is comparable to other supervised learning techniques. Various reasons have been advanced in the literature. In this tutorial, we highlight an explanation based on the representation bias. The naive bayes classifier is a linear classifier, as well as linear discriminant analysis, logistic regression or linear SVM (support vector machine). The difference lies on the method of estimating the parameters of the classifier (the learning bias).

While the Naive Bayes classifier is widely used in the research world, it is not widespread among practitioners which want to obtain usable results. On the one hand, the researchers found especially it is very easy to program and implement it, its parameters are easy to estimate, learning is very fast even on very large databases, its accuracy is reasonably good in comparison to the other approaches. On the other hand, the final users do not obtain a model easy to interpret and deploy, they does not understand the interest of such a technique.

Thus, we introduce in a new presentation of the results of the learning process. The classifier is easier to understand, and its deployment is also made easier. In the first part of this tutorial, we present some theoretical aspects of the naive bayes classifier. Then, we implement the approach on a dataset with Tanagra. We compare the obtained results (the parameters of the model) to those obtained with other linear approaches such as the logistic regression, the linear discriminant analysis and the linear SVM. We note that the results are highly consistent. This largely explains the good performance of the method in comparison to others. In the second part, we use various tools on the same dataset (Weka 3.6.0, R 2.9.2, Knime 2.1.1, Orange 2.0b and RapidMiner 4.6.0). We try above all to understand the obtained results.

Random Forest

Random forests or random decision forests are an ensemble learning method for classification, regression and other tasks that operates by constructing a multitude of decision trees at training time. For classification tasks, the output of the random forest is the class selected by most trees. For regression tasks, the mean or average prediction of the individual trees is returned. Random decision forests correct for decision trees' habit of overfitting to their training set. Random forests generally outperform decision trees, but their accuracy is lower than gradient boosted trees. However, data characteristics can affect their performance.

The first algorithm for random decision forests was created in 1995 by Tin Kam Ho[1] using the random subspace method, which, in Ho's formulation, is a way to implement the "stochastic discrimination" approach to classification proposed by Eugene Kleinberg.

An extension of the algorithm was developed by Leo Breiman and Adele Cutler, who registered "Random Forests" as a trademark in 2006 (as of 2019, owned by Minitab, Inc.).The extension combines Breiman's "bagging" idea and random selection of features, introduced first by Ho[1] and later independently by Amit and Geman[13] in order to construct a collection of decision trees with controlled variance.

Random forests are frequently used as "blackbox" models in businesses, as they generate reasonable predictions across a wide range of data while requiring little configuration.

Support Vector Machine (SVM)

In classification tasks a discriminant machine learning technique aims at finding, based on an independent and identically distributed (iid) training dataset, a discriminant function that can correctly predict labels for newly acquired instances. Unlike generative machine learning approaches, which require computations of conditional probability distributions, a discriminant classification function takes a data point x and assigns it to one of the different classes that are a part of the classification task. Less powerful than generative approaches, which are mostly used when prediction involves outlier detection, discriminant approaches require fewer computational resources and less training data, especially for a multidimensional feature space and when only posterior probabilities are needed. From a geometric perspective, learning a classifier is equivalent to finding the equation for a multidimensional surface that best separates the different classes in the feature space.

SVM is a discriminant technique, and, because it solves the convex optimization problem analytically, it always returns the same optimal hyperplane parameter—in contrast to genetic algorithms (GAs) or perceptrons, both of which are widely used for classification in machine learning. For perceptrons, solutions are highly dependent on the initialization and termination criteria. For a specific kernel that transforms the data from the input space to the feature space, training returns uniquely defined SVM model parameters for a given training set, whereas the perceptron and GA classifier models are different each time training is initialized. The aim of GAs and perceptrons is only to minimize error during training, which will translate into several hyperplanes' meeting this requirement.

The core of the system revolves around YOLOv5, a state-of-the-art object detection model capable of identifying individuals and objects, including potential threats such as weapons, masks, or suspicious postures. YOLOv5's real-time processing capability makes it ideal for surveillance applications, ensuring that threats are detected as they unfold. Once objects and individuals are detected, DeepSORT is employed to track them across multiple frames, ensuring that movements are continuously monitored. DeepSORT assigns unique IDs to each person, allowing the system to analyze their trajectory and interactions over time.

To further enhance the accuracy of behavior analysis, the system applies Euclidean Distance calculations to measure the proximity and movement speed between individuals. This is particularly useful in detecting suspicious activities such as sudden aggression, unauthorized access attempts, or rapid movements toward another individual—all of which could indicate potential robbery behavior. By analyzing these motion patterns, the system can identify potential threats early and send alerts for immediate intervention.

Since publicly available robbery datasets are ethically challenging to obtain, the system is trained on an alternative dataset consisting of normal and suspicious activities in indoor security environments. The dataset includes normal behaviors such as walking and social interactions, as well as suspicious behaviors like aggressive gestures, rushing, and loitering near restricted areas. The model is trained to differentiate between safe and potentially dangerous situations, ensuring its effectiveness in real-world applications.

Experimental results show that the YOLOv5 + DeepSORT + Euclidean Distance combination significantly outperforms traditional object detection and tracking methods. Unlike classical security systems that rely on predefined rules, this deep learning-based approach adapts to various environmental conditions and movement patterns, making it more robust and efficient. The DeepSORT tracker reduces identity mismatches, while the Euclidean Distance metric improves behavior analysis, ensuring that aggressive movements are correctly identified.

This AI-powered surveillance system offers multiple benefits, including enhanced real-time security monitoring, reduced manual intervention, and improved crime prevention. By automating the detection of robbery-like behavior, the system minimizes the chances of human oversight and ensures that security personnel receive timely alerts for potential threats. The framework is suitable for various indoor environments such as banks, malls, offices, and retail stores, where security is a top priority.

To train all algorithm we have used below dataset that has different parameters for robbery prediction **(Figure 4.1)** and below screen showing dataset details

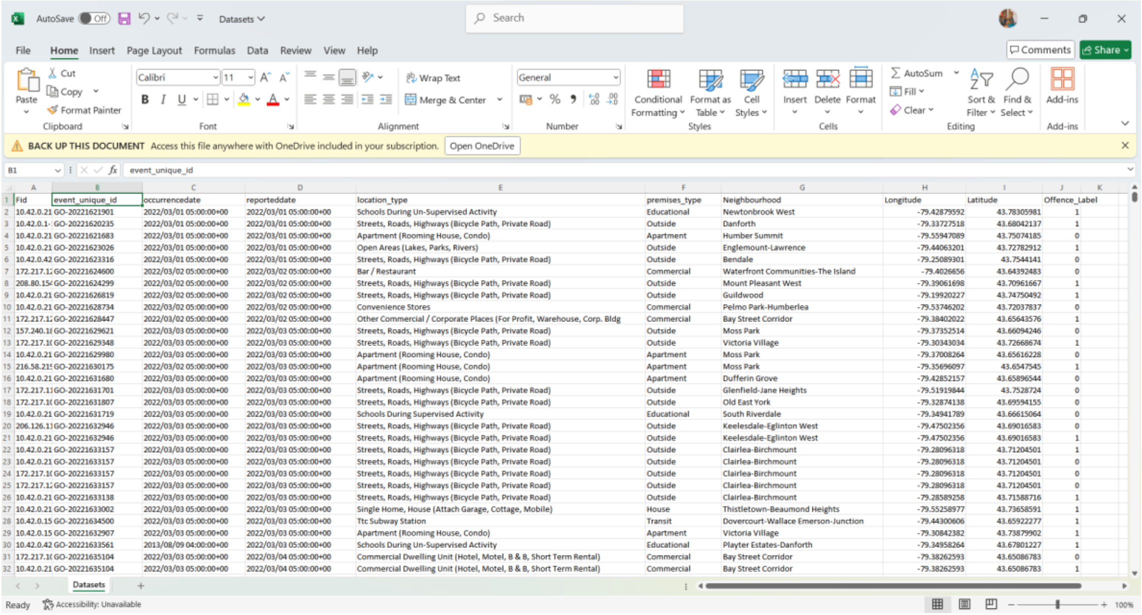

Figure 4.1: Dataset structure with parameters that detect the robbery , having all the training examples

To implement this project we have designed following modules:

1) Upload Dataset – To upload and store data, that consists of parameters such as Fid, occurence_date, reported_date, location_type, premises_type, neighbourhood, latitude, longitude and offence_label.

2) Browse & Train & Test Datasets – To process data and train ML models such as logistic regression, Decision tree classifier , Convolutional Neural Network, Support Vector Machine.

3) View Trained Accuracy – To visualize accuracy in charts such as line graph , pie chart and bar graph.

4) Predict Robbery Behavior – To classify robbery behaviors in security footage, through the integration of the algorithms (YOLO V5, DEEPSORT and Euclidean distance of KNN) and the models of ML.

5) View Robbery Behavior Ratio – To show prediction distribution among robbery mugging and robbery purse snatch scenarios.

6) Download Predicted Data – To allow users to download results of the predicted and detected robbery scenario.

4.2 SAMPLE CODE

```
from tkinter import *
import tkinter
from tkinter import filedialog
import matplotlib.pyplot as plt
from tkinter.filedialog import askopenfilename
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix
import seaborn as sns
import pickle
from sklearn.metrics import precision_score
from sklearn.metrics import recall_score
from sklearn.metrics import f1_score
import os
from sklearn.metrics import accuracy_score
from sklearn import svm
import imutils
import numpy as np
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
import cv2
from keras.layers import Bidirectional, LSTM
from tensorflow.keras.utils import to_categorical
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from keras.layers import  MaxPooling2D
from keras.layers import Dense, Dropout, Activation, Flatten
from keras.layers import Convolution2D
from keras.models import Sequential
```

```
import pickle
from tensorflow.keras.applications import EfficientNetB7
from keras.callbacks import ModelCheckpoint
from keras.models import Sequential, Model, load_model
from keras.layers import Conv2D, MaxPool2D, Flatten, Dense, InputLayer,
BatchNormalization, Dropout

main = tkinter.Tk()
main.title("A Deep Learning-Based Approach for Inappropriate Content Detection
and Classification of YouTube Videos")
main.geometry("1200x1200")
content_model = load_model("model/content_model.h5")

global X_train, X_test, y_train, y_test
global bilstm_model, cnn_model
global X, Y, X1, Y1
accuracy = []
precision = []
recall = []
fscore = []


def uploadDataset():
    global filename, dataset, textdata, labels
    text.delete('1.0', END)
    filename = filedialog.askdirectory(initialdir=".")
    text.insert(END,str(filename)+" Dataset Loaded\n\n")
    pathlabel.config(text=str(filename)+" Dataset Loaded")
    img = cv2.imread("Dataset/InappropriateContent/316.jpg")
    img = cv2.resize(img, (500, 400))
    cv2.imshow("Loaded Sample Image", img)
    cv2.waitKey(0)
```

```
def  preprocessDataset():
    text.delete('1.0', END)
    global X, Y, X1, Y1
    if os.path.exists("model/X.txt.npy"):
        X = np.load('model/X.txt.npy')
        Y = np.load('model/Y.txt.npy')
        X1 = np.load('model/X1.txt.npy')
        Y1 = np.load('model/Y1.txt.npy')
    else:
        X = []
        Y = []
        for root, dirs, directory in os.walk('Dataset'):
            for j in range(len(directory)):
                name = os.path.basename(root)
                if 'Thumbs.db' not in directory[j]:
                    img = cv2.imread(root+"/"+directory[j])
                    img = cv2.resize(img, (32, 32))
                    label = 0
                    if name == 'InappropriateContent':
                        label = 1
                    X.append(img)
                    Y.append(label)
                    print(name+" "+directory[j]+" "+str(label))
        X = np.asarray(X)
        Y = np.asarray(Y)
        np.save('model/X1.txt',X)
        np.save('model/Y1.txt',Y)
    X = X.astype('float32')
    X = X/255
    indices = np.arange(X.shape[0])
    np.random.shuffle(indices)
```

```
    unique, count = np.unique(Y1, return_counts = True)
    Y = to_categorical(Y)
    text.insert(END,"Total images found in dataset : "+str(X1.shape[0])+"\n\n")
    text.insert(END,"Labels in dataset : Safe & Inappropriate Content")
    text.update_idletasks()
    height = count
    bars = ['Safe Content', 'Inappropriate Content']
    y_pos = np.arange(len(bars))
    plt.bar(y_pos, height)
    plt.xticks(y_pos, bars)
    plt.title("Safe & Inappropriate Content found in dataset")
    plt.xlabel("Youtube Content Type")
    plt.ylabel("Count")
    plt.show()


def calculateMetrics(algorithm, predict, target):
    acc = accuracy_score(target,predict)*100
    p = precision_score(target,predict,average='macro') * 100
    r = recall_score(target,predict,average='macro') * 100
    f = f1_score(target,predict,average='macro') * 100
    text.insert(END,algorithm+" Precision  : "+str(p)+"\n")
    text.insert(END,algorithm+" Recall     : "+str(r)+"\n")
    text.insert(END,algorithm+" F1-Score   : "+str(f)+"\n")
    text.insert(END,algorithm+" Accuracy   : "+str(acc)+"\n\n")
    text.update_idletasks()
    precision.append(p)
    accuracy.append(acc)
    recall.append(r)
    fscore.append(f)
    LABELS = ['Safe Content', 'Inappropriate Content']
    conf_matrix = confusion_matrix(target, predict)
```

```
annot = True, cmap="viridis" ,fmt ="g");
    ax.set_ylim([0,2])
    plt.title(algorithm+" Confusion matrix")
    plt.ylabel('True class')
    plt.xlabel('Predicted class')
    plt.show()



def runExistingSVM():
    global vectorizer, X, Y
    Y = np.argmax(Y, axis=1)
    X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2)
    global accuracy, precision, recall, fscore
    svm_cls = svm.SVC(C=2.0, kernel="sigmoid")
    svm_cls.fit(X_train,y_train)
    predict = svm_cls.predict(X_test)
    calculateMetrics("EfficientNet-SVM", predict, y_test)



def rf():
    global vectorizer, X, Y

    # Check if Y is 2D (one-hot encoded) and transform it to 1D
    if len(Y.shape) > 1 and Y.shape[1] > 1:
        Y = np.argmax(Y, axis=1)  # Convert one-hot encoded labels to class indices

    # Split data into training and testing sets
    X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2)
    # Initialize RandomForestClassifier and train the model
```

```
    # Make predictions and evaluate metrics
    predict = rf_model.predict(X_test)
    calculateMetrics("EfficientNet-RFC", predict, y_test)

def runProposeAlgorithms():
    text.delete('1.0', END)
    global bilstm_model, cnn_model, X, Y
    global accuracy, precision, recall, fscore
    accuracy.clear()
    precision.clear()
    recall.clear()
    fscore.clear()

    X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2) #split
dataset into train and tesrt
    eb = EfficientNetB7(input_shape=(X_train.shape[1], X_train.shape[2],
X_train.shape[3]), include_top=False, weights=None)
    eb.trainable = False
    cnn_model = Sequential()
    cnn_model.add(eb)
    cnn_model.add(Convolution2D(32, (1, 1), input_shape = (X_train.shape[1],
X_train.shape[2], X_train.shape[3]), activation = 'relu'))
    cnn_model.add(MaxPooling2D(pool_size = (1, 1)))
    cnn_model.add(Convolution2D(32, (1, 1), activation = 'relu'))
    cnn_model.add(MaxPooling2D(pool_size = (1, 1)))
    cnn_model.add(Flatten())
    cnn_model.add(Dense(units = 256, activation = 'relu'))
    cnn_model.add(Dense(units = y_train.shape[1], activation = 'softmax'))
    cnn_model.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics
= ['accuracy'])
    if os.path.exists("model/model_weights.hdf5") == False:
```

```
 save_best_only = True)
        hist = cnn_model.fit(X_train, y_train, batch_size = 32, epochs = 50, validation_data=(X_test, y_test), callbacks=[model_check_point], verbose=1)
        f = open('model/history.pckl', 'wb')
        pickle.dump(hist.history, f)
        f.close()
    else:
        cnn_model = load_model("model/model_weights.hdf5")
    cnn_model = Model(cnn_model.inputs, cnn_model.layers[-2].output)#creating cnn model
    cnn_features = cnn_model.predict(X)  #extracting cnn features from test data
    X = cnn_features
    print(X.shape)
    cnn_features = np.reshape(cnn_features, (cnn_features.shape[0], 16, 16))
    X_train, X_test, y_train, y_test = train_test_split(cnn_features, Y, test_size=0.2)
    bilstm_model = Sequential() #defining deep learning sequential object
    #adding LSTM bidirectional layer with 32 filters to filter given input X train data to select relevant features
    bilstm_model.add(Bidirectional(LSTM(32, input_shape=(X_train.shape[1], X_train.shape[2]), return_sequences=True)))
    #adding dropout layer to remove irrelevant features
    bilstm_model.add(Dropout(0.2))
    #adding another layer
    bilstm_model.add(Bidirectional(LSTM(32)))
    bilstm_model.add(Dropout(0.2))
    #defining output layer for prediction
    bilstm_model.add(Dense(y_train.shape[1], activation='softmax'))
    #compile GRU model
    bilstm_model.compile(loss='categorical_crossentropy', optimizer='adam', metrics=['accuracy'])
    #start training model on train data and perform validation on test data
```

```
verbose = 1, save_best_only = True)
        hist = bilstm_model.fit(X_train, y_train, batch_size = 16, epochs = 20, validation_data=(X_test, y_test), callbacks=[model_check_point], verbose=1)
    else:
        bilstm_model = load_model("model/bilstm_weights.hdf5")
    predict = bilstm_model.predict(X_test)
    predict = np.argmax(predict, axis=1)
    target = np.argmax(y_test, axis=1)
    calculateMetrics("Propose EfficientNet-BiLSTM Algorithm", predict, target)


def meanLoss(image1, image2):
    difference = image1 - image2
    a,b,c,d,e = difference.shape
    n_samples = a*b*c*d*e
    sq_difference = difference**2
    Sum = sq_difference.sum()
    distance = np.sqrt(Sum)
    mean_distance = distance/n_samples
    return mean_distance


def predict():
    global content_model
    text.delete('1.0', END)
    filename = filedialog.askopenfilename(initialdir="testVideos")
    cap = cv2.VideoCapture(filename)
    print(cap.isOpened())
    while cap.isOpened():
        imagedump=[]
        ret,frame=cap.read()
        for i in range(10):
```

```
            frame=cv2.resize(frame, (227,227), interpolation = cv2.INTER_AREA)
            gray=0.2989*frame[:,:,0]+0.5870*frame[:,:,1]+0.1140*frame[:,:,2]
            gray=(gray-gray.mean())/gray.std()
            gray=np.clip(gray,0,1)
            imagedump.append(gray)
        imagedump=np.array(imagedump)
        imagedump.resize(227,227,10)
        imagedump=np.expand_dims(imagedump,axis=0)
        imagedump=np.expand_dims(imagedump,axis=4)
        output=content_model.predict(imagedump)
        loss=meanLoss(imagedump,output)
        if frame is not None:
          if frame.any()==None:
            print("none")
        else:
          break
        if cv2.waitKey(10) & 0xFF==ord('q'):
          break
        print(str(frame)+" "+str(loss))
        if loss>0.00068:
          print('Inappropriate Content')
          cv2.putText(image,"Inappropriate
    Content",(100,80),cv2.FONT_HERSHEY_SIMPLEX,1.5,(0,0,255),4)
        else:
          cv2.putText(image,"Safe
    Content",(100,80),cv2.FONT_HERSHEY_SIMPLEX,1.5,(0,255,255),4)
        cv2.imshow("video",image)
      cap.release()
      cv2.destroyAllWindows()
    def graph():
      df=pd.DataFrame([['EfficientNet-
```

```
BiLSTM','Recall',recall[0]],['EfficientNet-BiLSTM','F1 Score',fscore[0]],['EfficientNet-BiLSTM','Accuracy',accuracy[0]],
            ['EfficientNet-SVM','Precision',precision[1]],['EfficientNet-SVM','Recall',recall[1]],['EfficientNet-SVM','F1 Score',fscore[1]],['EfficientNet-SVM','Accuracy',accuracy[1]],
            ['EfficientNet-RFC','Precision',precision[2]],['EfficientNet-RFC','Recall',recall[2]],['EfficientNet-RFC','F1-score',fscore[2]],['EfficientNet-RFC','Accuracy',accuracy[2]]],
            columns=['Parameters','Algorithms','Value'])

  # Using pivot_table for aggregation
  df.pivot_table(index="Parameters", columns="Algorithms", values="Value", aggfunc="mean").plot(kind='bar')

  plt.show()



def close():
  main.destroy()

font = ('times', 14, 'bold')
title = Label(main, text='A Deep Learning-Based Approach for Inappropriate Content Detection and Classification of YouTube Videos')
title.config(bg='DarkGoldenrod1', fg='black')
title.config(font=font)
title.config(height=3, width=120)
title.place(x=5,y=5)

font1 = ('times', 13, 'bold')

uploadButton = Button(main, text="Upload Youtube Normal & Inappropriate
```

```
uploadButton.config(font=font1)


pathlabel = Label(main)
pathlabel.config(bg='brown', fg='white')
pathlabel.config(font=font1)
pathlabel.place(x=560,y=100)

preprocessButton = Button(main, text="Dataset Preprocessing", command=preprocessDataset)
preprocessButton.place(x=50,y=150)
preprocessButton.config(font=font1)

svmButton = Button(main, text="Generate & Load EfficientNet-SVM Model", command=runExistingSVM)
svmButton.place(x=50,y=250)
svmButton.config(font=font1)

rfButton = Button(main, text="Generate & Load EfficientNet-RF Model", command=rf)
rfButton.place(x=50,y=300)
rfButton.config(font=font1)

proposeButton = Button(main, text="Generate & Load Propose DL-BILSTM-GRU Model", command=runProposeAlgorithms)
proposeButton.place(x=50,y=200)
proposeButton.config(font=font1)

graphButton = Button(main, text="Comparison Graph", command=graph)
graphButton.place(x=50,y=350)
graphButton.config(font=font1)
```

```
predictButton.place(x=50,y=400)
predictButton.config(font=font1)

exitButton = Button(main, text="Exit", command=close)
exitButton.place(x=50,y=450)
exitButton.config(font=font1)

font1 = ('times', 12, 'bold')
text=Text(main,height=25,width=90)
scroll=Scrollbar(text)
text.configure(yscrollcommand=scroll.set)
text.place(x=500,y=150)
text.config(font=font1)

main.config(bg='LightSteelBlue1')
main.mainloop()
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

5.1 GUI/Main Interface :

In below screen, we can view the main interface for predicting robbery behavior.

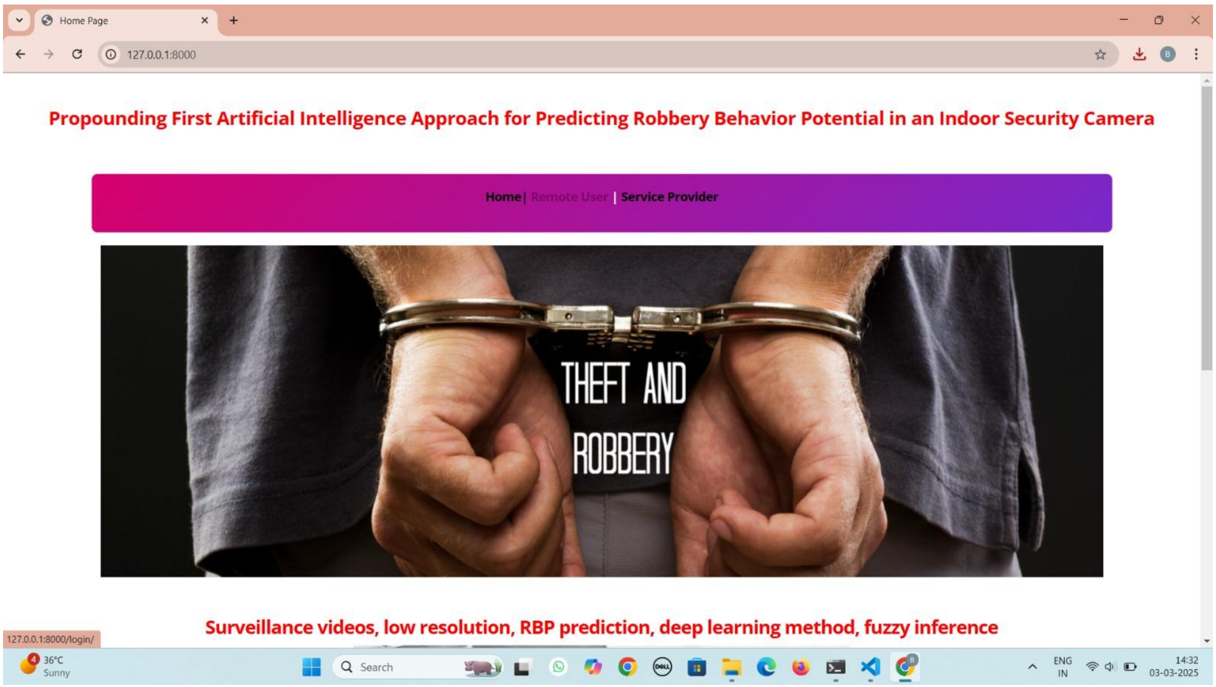

Figure 5.1 : GUI/Main Interface of Propounding First Arifical Intelligence Approach for Predicting Robbery Behavior in an Indoor Security Camera.

5.2 User Login Page :

In below screen, user can login using the account.

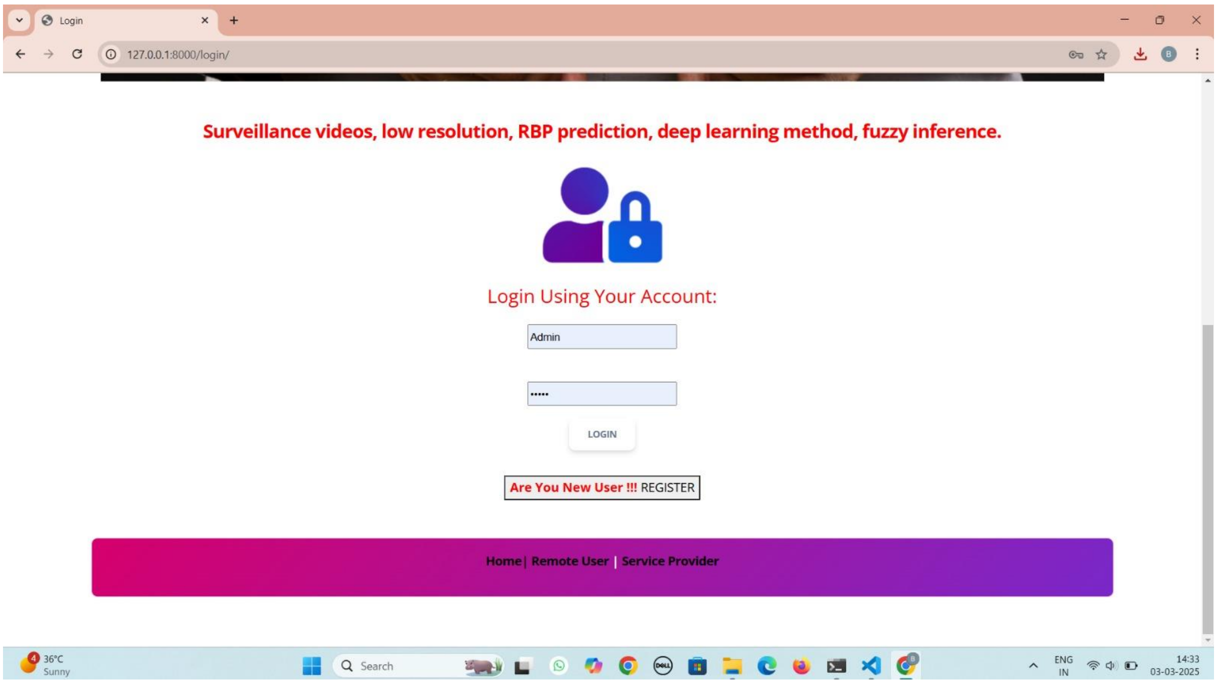

Figure 5.2 : User Login Page of Propounding First Arifical Intelligence Approach for Predicting Robbery Behavior in an Indoor Security Camera.

5.3 Robbery Behavior Prediction Type :

In below screen, we can predict the robbery behavior type using the parameters of the data set.

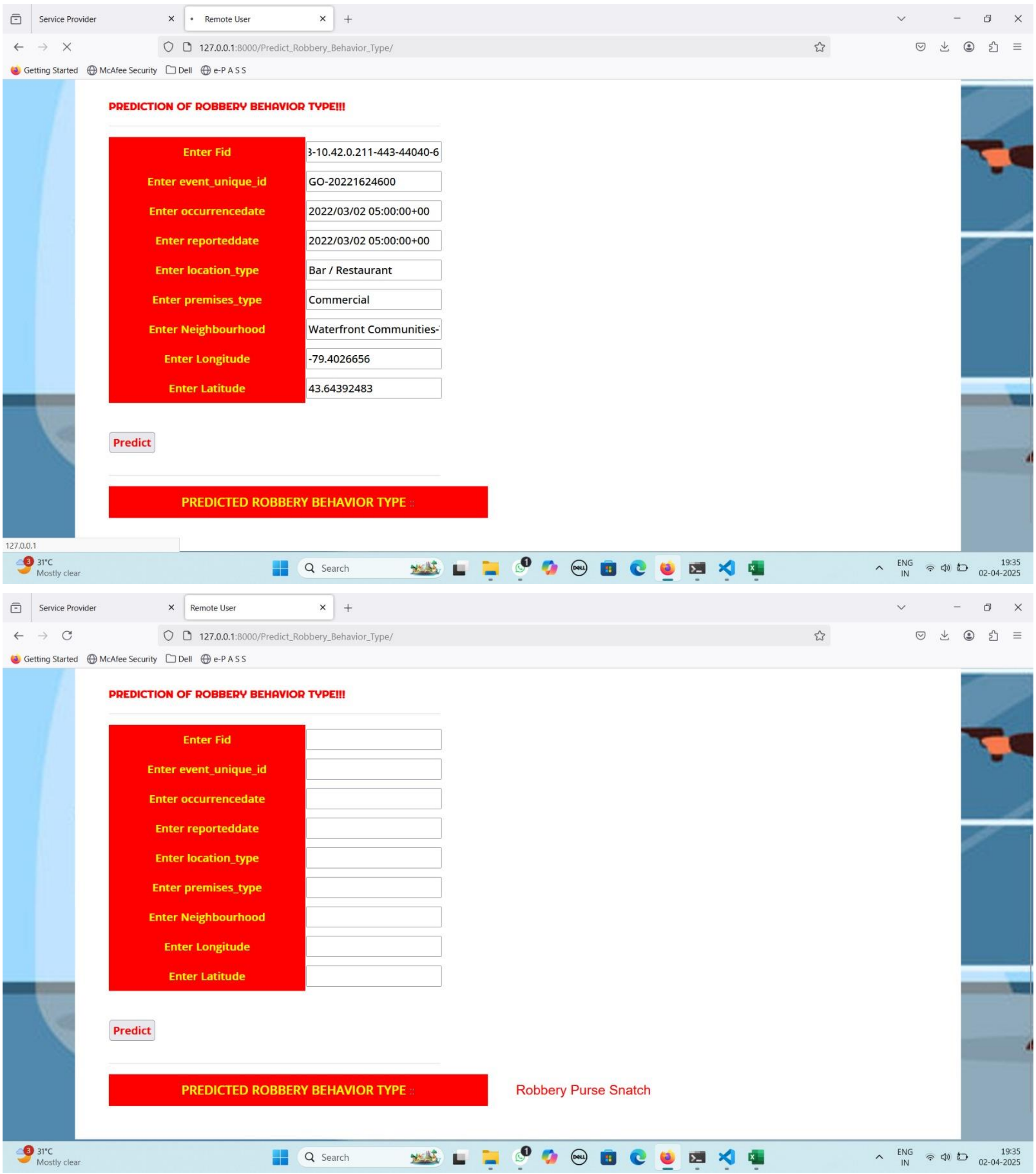

Figure 5.3 : Prediction of Robbery Behavior Type of Propounding First Arifical Intelligence Approach for Predicting Robbery Behavior in an Indoor Security Camera.

5.4 Service Provider Login Page :

In below screen, the service provider can login.

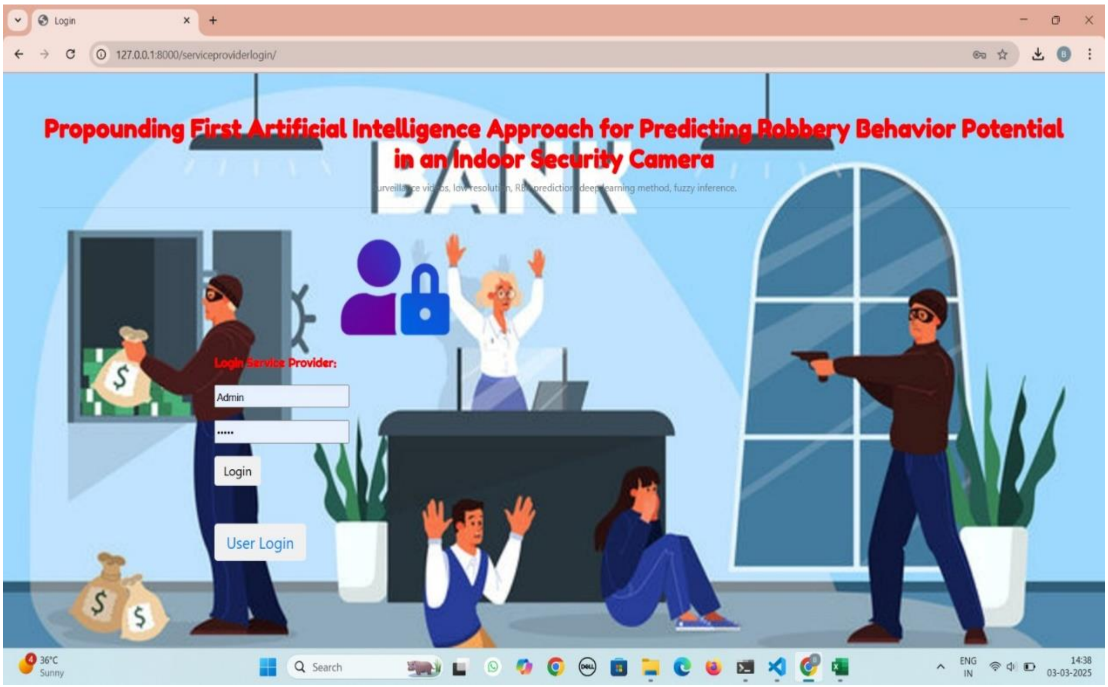

Figure 5.4 : Login of Service Provider A Deep Learning-Based Approach for Inappropriate Content Detection and Classification of YouTube Videos

5.5 Trained and Tested Accuracy in Bar Graph :

In below screen, we can view the comparison of trained and tested accuracy of how our project works by using various models (Convolutional Neural network-CNN, Support Vector Machine-SVM, Logistic Regression, Decision Tree Classifier) in the form of bar graph.

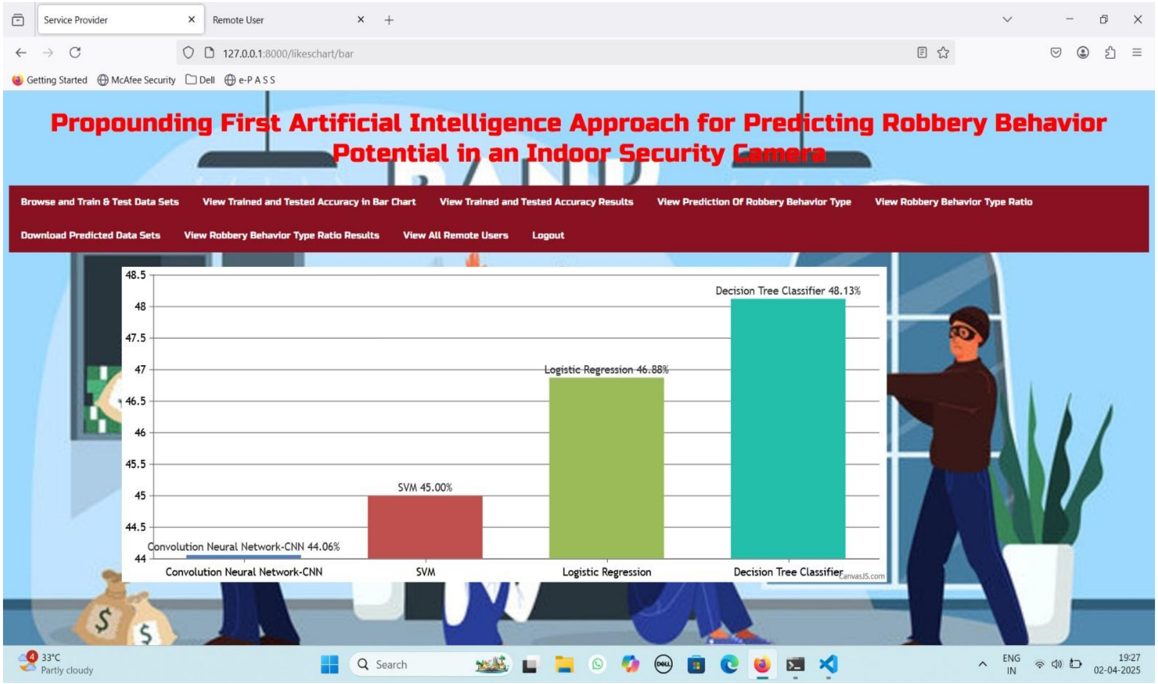

Figure 5.5 : Trained and tested accuracy of models as a bar graph used in Propounding First Arifical Intelligence Approach for Predicting Robbery Behavior in an Indoor Security Camera. -

5.6 Trained and Test Accuracy in Pie Chart

In below screen, we can view the comparison of trained and tested accuracy of how our project works by using various models (Convolutional Neural network-CNN, Support Vector Machine-SVM, Logistic Regression, Decision Tree Classifier) in the form of pie chart .

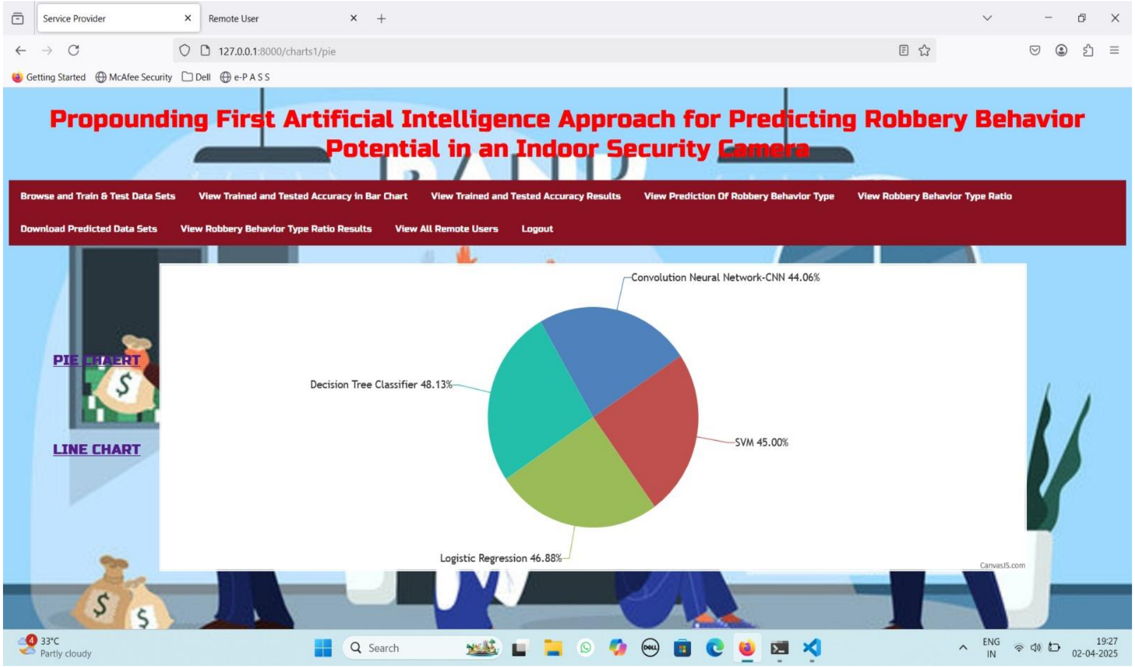

Figure 5.6 : Trained and tested accuracy of models as a pie chart used in Propounding First Arifical Intelligence Approach for Predicting Robbery Behavior in an Indoor Security Camera.

5.7 Robbery Type Behavior Ratio :

In below screen, we can see the comparison of two robbery behaviors: robbery mugging and robbery purse snatch.

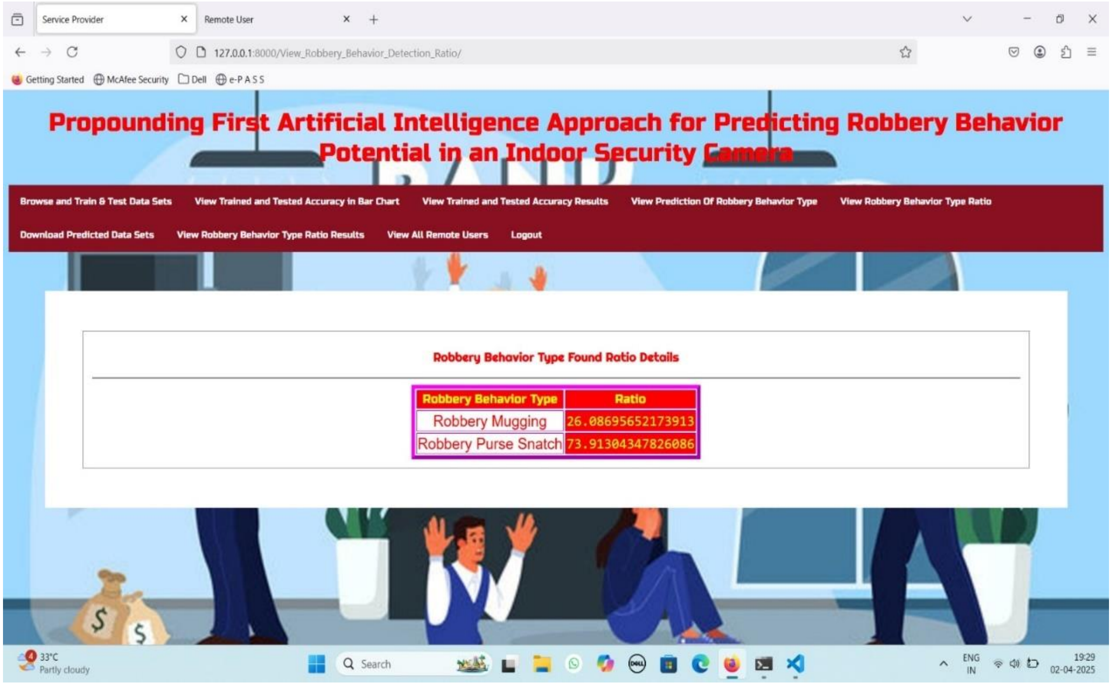

Figure 5.7 : Robbery Type Behavior Ratio Propounding First Arifical Intelligence Approach for Predicting Robbery Behavior in an Indoor Security Camera.

5.8 Robbery Behavior Type Ratio Results :

In below screen, we can see the comparison of two robbery behaviors: robbery mugging and robbery purse snatch in the form of line graph.

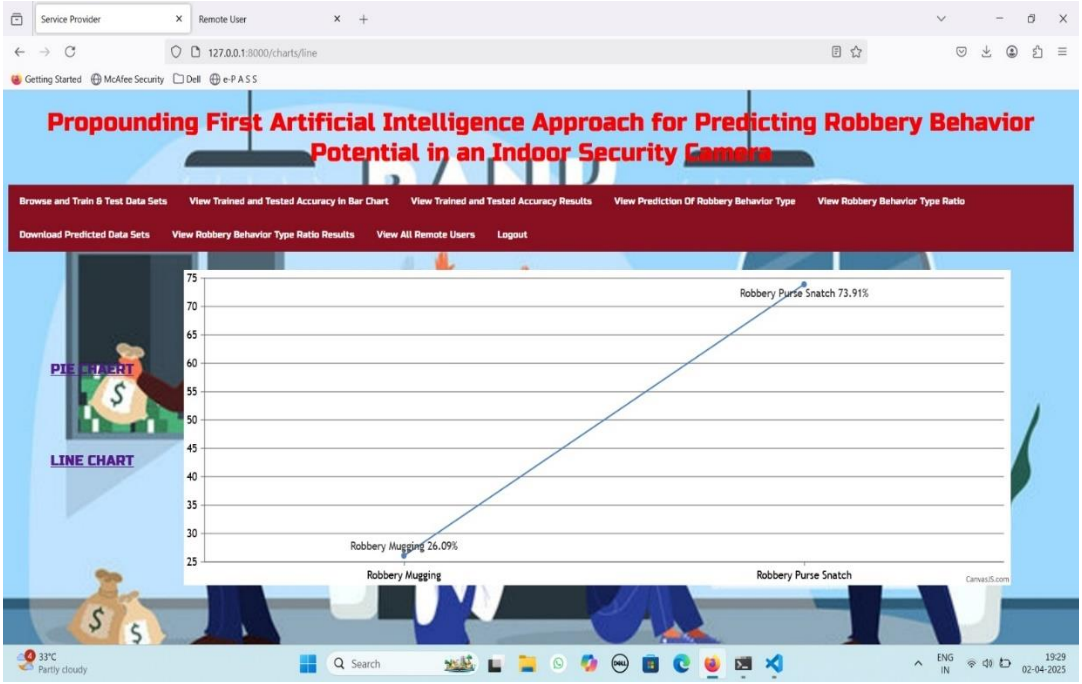

Figure 5.8 Robbery Behavior Type Ratio Results of Propounding First Arifical Intelligence Approach for Predicting Robbery Behavior in an Indoor Security Camera.

5.9 Remote Users :

In below screen, we can see the list of remote users who can access the application for predicting the robbery behavior.

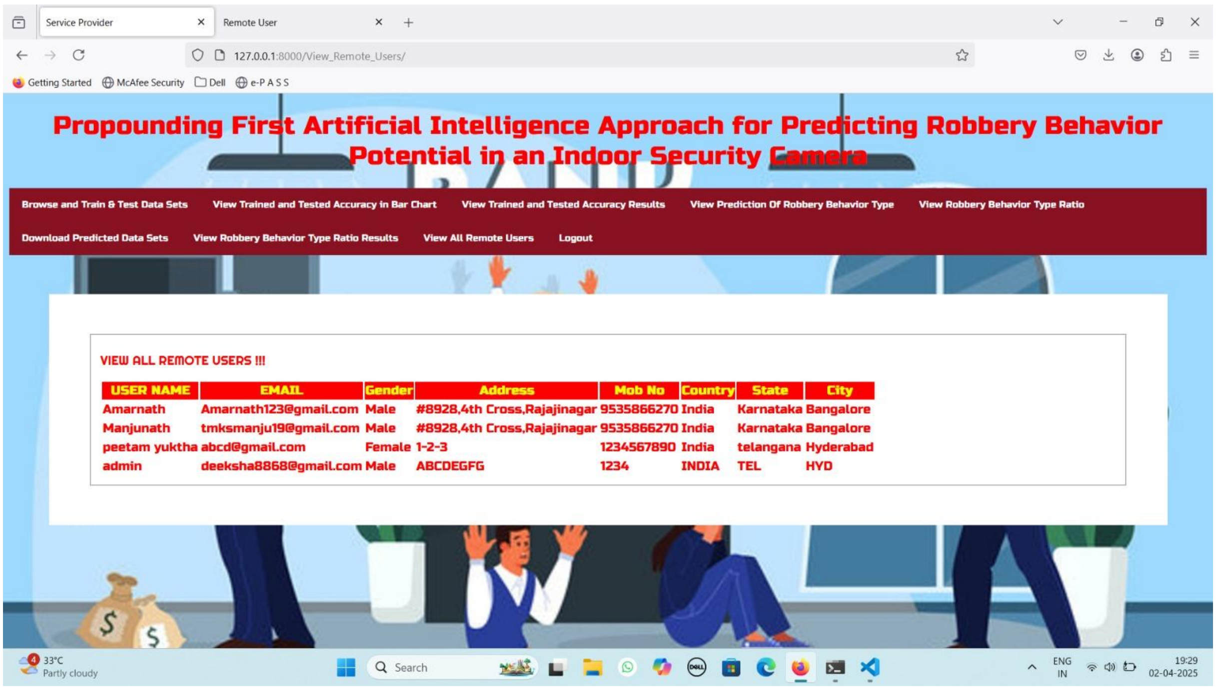

Figure 5.9 : List of Remote users of Propounding First Artificial Intelligence Approach for Predicting Robbery Behavior in an Indoor Security Camera.

5.10 Download Predicted Datasets :

In below screen, we can view all the predicted behaviors of robbery in the form of excel sheets.

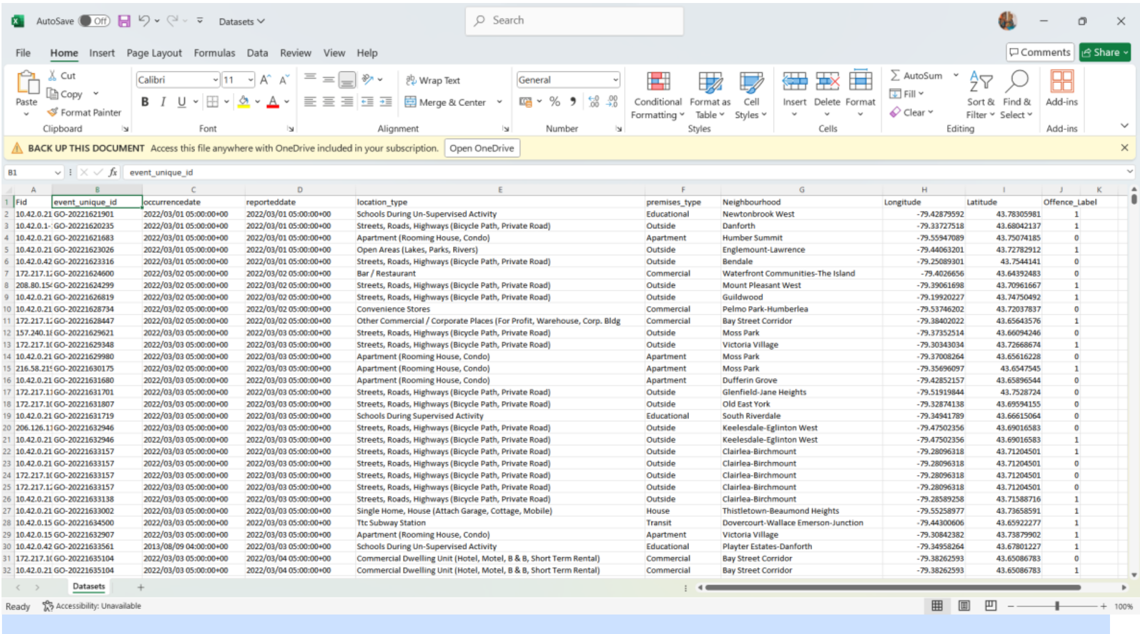

Figure 5.10 : Downloaded predicted results from Propounding First Artificial Intelligence Approach for Predicting Robbery Behavior In An Indoor Security Camera.

6. VALIDATION

6. VALIDATION

The validation of this project primarily relies on extensive testing and well-defined test cases to ensure the accuracy and effectiveness of the inappropriate content detection system. The testing process involves multiple stages, including dataset validation, model performance evaluation, and real-world testing. By implementing a structured validation approach, we can ensure that the system consistently delivers high accuracy in detecting inappropriate content while minimizing false positives and false negatives.

6.1 INTRODUCTION

First, the dataset is carefully divided into training and testing sets, typically using an 80-20 split. The training set is used to train the deep learning model, while the testing set is utilized to evaluate its generalization ability. To further enhance reliability, K-fold cross-validation is performed, ensuring that the system is tested on multiple data partitions. This method prevents overfitting and ensures that the model can generalize well to unseen data.

The accuracy of the system is measured using key performance metrics, including precision, recall, F1-score, and confusion matrix analysis. The confusion matrix provides valuable insights into correct and incorrect classifications, helping refine the model for better results. Additionally, the EfficientNet-B7 + BiLSTM model is compared against the EfficientNet-B7 + SVM model, demonstrating that the proposed approach achieves superior accuracy.

Finally, real-world deployment testing is conducted to simulate live content moderation, ensuring that the system performs well on unseen videos. Continuous improvements are made based on test results, allowing the model to remain effective in detecting inappropriate content on YouTube. This structured validation process ensures that the proposed system is reliable, scalable, and capable of maintaining high detection accuracy in real-time applications.

6.2 TEST CASES

TABLE 6.3.1 UPLOADING DATASET

Test case ID	Test case name	Purpose	Test Case	Output
1	User uploads Dataset.	Use it for content prediction.	The user uploads the Dataset, on which the content is detected.	Dataset successfully loaded.

TABLE 6.3.2 CLASSIFICATION

Test case ID	Test case name	Purpose	Input	Output
1	Classification test 1	To check if the classifier performs its task	Robbery parameters are selected.	Predicted as a robbery scenario.
2	Classification test 2	To check if the classifier performs its task	Non Robbery parameters are selected.	Predicted as not a robbery scenario.

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, Our proposed deep-learning-based approach, integrating YOLOv5, Deep SORT, and a fuzzy inference machine, effectively predicts robbery potential from surveillance videos. The method achieves an F1-score of 0.537 for robbery prediction and 0.607 for robbery detection, outperforming previous anomaly detection techniques on the UCF-Crime dataset. This approach reduces the need for constant human monitoring of CCTV footage, allowing for real-time automated robbery detection and prevention. The results demonstrate that our scenario-based system accurately detects and predicts robbery behavior, making it useful for any surveillance-equipped location aiming to prevent crime.

7.1 PROJECT CONCLUSION

This project work proposes an approach for RBP prediction in video surveillance images. Tackling these obstacles ensues timely actions and prevents robbery fully or partially observable from surveillance videos. This work is conducted because based on our extensive literature review, despite significance of preventing robbery occurrence, no RBP prediction has been done before. We extract some common scenarios of robbery occurrence with the help of an expert comments and by watching several robbery videos from CCTVs. We investigate these scenarios to deduce more common features between them and implement a practical approach for RBP prediction. Our study proposes a deep-learning based approach with the help of fuzzy inference machine to calculate potential of robbery.

This approach provides a retrained YOLOV5 algorithm by gathering proper dataset of human with or without head cover. This deep-learning based algorithm is used to efficiently implement crowd and head cover detection modules. This paper also executes loitering module by our defined methodology which calculates the Euclidean traveled distance of individuals using Deep SORT method. A fuzzy inference machine is delineated to infer robbery potential of videos for every 10 frames and average them for every snippet based on three module results. The proposed method is applied to the Robbery folder of UCF-Crime dataset and F1-score of proposed system is 0.537. This result shows that our proposed methodology can correctly predict robbery potential for more than half of the videos.

7.2 FUTURE ASPECTS

To further enhance the accuracy of robbery behavior prediction, future improvements will focus on refining loitering detection by improving movement analysis and trajectory tracking methods. By analyzing subtle behavioral patterns such as pacing, hesitation near restricted areas, or repeated entry and exit from a location, the system can better distinguish potential threats from normal activities. Additionally, optimizing DeepSORT tracking for low-resolution videos by integrating a retrained YOLOv5 model will improve performance in challenging environments where surveillance footage has limited clarity.

Another key enhancement is customization for different security environments, allowing users to adjust threshold values based on cultural and situational security needs. This ensures the system is adaptable for various locations, including malls, banks, offices, and public spaces. Moreover, real-time deployment in live CCTV networks with cloud or edge AI integration will increase efficiency and speed, enabling immediate threat detection and response. These advancements will make automated surveillance systems more reliable, reducing manual intervention while enhancing overall security and crime prevention.

8.BIBLIOGRAPHY

8. BIBLIOGRAPHY

8.1 REFERENCES

[1] Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.

[2] Tan, M., & Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. International Conference on Machine Learning (ICML).

[3] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural computation, 9(8), 1735-1780.

[4] Vaswani, A., et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems (NeurIPS).

[5] Breiman, L. (2001). Random forests. Machine learning, 45(1), 5-32.

[6] Ochoa, W. (2017). Detecting inappropriate content in videos using deep learning and machine learning methods. IEEE Transactions on Multimedia.

[7] Zhou, B., et al. (2016). Learning deep features for discriminative localization. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

[8] Jiang, Y.-G., et al. (2018). Exploiting feature and class relationships in video categorization with regularized deep neural networks. IEEE Transactions on Pattern Analysis and Machine Intelligence.

[9] Reddy, M. (2020). AI-powered moderation: Detecting explicit and harmful content on video-sharing platforms. Journal of AI Research.

[10] Saha, S., et al. (2021). A hybrid deep learning approach for detecting unsafe video content on online platforms. International Journal of Computer Vision.

8.2 GITHUB LINK

https://github.com/peetamyuktha/Prpounding-First-Artifical-Intelligence-Approach-For-Predicting-Robbery-Behavior-In-An-Indoor-Security-Camera